

03-) 50h ; $c = \frac{4\text{KJ}}{\text{kg}\cdot\text{K}}$; $d = 5 \cdot \frac{\text{kg}}{\text{L}}$; $\Delta t = 3\text{h} = 60 \cdot 60 = 3600\text{s}$

$P = 50\text{W}$

$P = \frac{E}{\Delta t} \rightarrow E = P \cdot \Delta t \rightarrow E = 50 \cdot 3600 = 5 \cdot 36 \cdot 10^3 = 180 \cdot 10^3 \text{ J} = 180\text{KJ}$

$d = \frac{m}{V} \rightarrow \frac{5}{50} = \frac{m}{50} \rightarrow m = 50\text{kg}$

$Q = m \cdot c \cdot \Delta T \rightarrow 180 = \frac{50 \cdot 4 \cdot \Delta T}{200} \rightarrow \Delta T = \frac{180}{200} = \frac{18}{20} < 1$

$\rightarrow 0,9^\circ\text{C}$

alternativa "c"

04-) $Q = m \cdot c \cdot \Delta T$

Água $\rightarrow Q_A = m_A \cdot c_A \cdot \Delta T_A \rightarrow \Delta T_A = \frac{Q_A}{m_A \cdot c_A}$

Concreto $\rightarrow Q_C = m_C \cdot c_C \cdot \Delta T_C \rightarrow \Delta T_C = \frac{Q_C}{m_C \cdot c_C}$

$\frac{\Delta T_C}{\Delta T_A} = \frac{\frac{Q_C}{m_C \cdot c_C}}{\frac{Q_A}{m_A \cdot c_A}} = \frac{Q_C}{m_C \cdot c_C} \cdot \frac{m_A \cdot c_A}{Q_A}$; $Q_A = Q_C$

$d = \frac{m}{V} \rightarrow m = d \cdot V \rightarrow \frac{\Delta T_C}{\Delta T_A} = \frac{m_A \cdot c_A}{m_C \cdot c_C} = \frac{d_A \cdot V \cdot c_A}{d_C \cdot V \cdot c_C} = \frac{d_A \cdot c_A}{d_C \cdot c_C} = \frac{1000 \cdot 4,2}{2500 \cdot 0,8}$

$\rightarrow \frac{4,2 \cdot 10}{25 \cdot 0,8} = \frac{4,2 \cdot 5 \cdot 2}{5 \cdot 5 \cdot 0,8} = \frac{4,2 \cdot 2}{5 \cdot 0,8} = \frac{4,2 \cdot 2}{4}$

$\frac{4,2 \cdot 2}{2 \cdot 2} = \frac{4,2}{2} = 2,1$ alternativa "b"

05-) $V = 220 \text{ V}$; $c = 4,2 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$; $d = 1000 \frac{\text{g}}{\text{L}} = \frac{1 \text{ kg}}{\text{L}}$

$$P = \frac{E}{\Delta t} = \frac{[J]}{[s]}$$

$Q = m \cdot c \cdot \Delta T \rightarrow$ pior caso: Superquente, a vazão compensa o ΔT , então $\Delta T = 15^\circ\text{C}$

ou $\Delta T = 30^\circ\text{C}$ o resultado é o mesmo.

; Vazão (V_A) $\rightarrow V_A = \frac{\text{volume (V)}}{\text{tempo (t)}} = \frac{[\text{Litros}]}{[\text{minutos}]}$;

$$d = \frac{m}{V} \rightarrow V = \frac{m}{d} \quad \rightarrow \quad V_A = \frac{\frac{m}{d}}{\Delta t} \rightarrow V_A = \frac{m}{d \cdot \Delta t}$$

$$V_A \cdot d = \frac{m}{\Delta t}$$

$$Q = m \cdot c \cdot \Delta T$$

$$P = \frac{Q}{s} = V_A \cdot d \cdot c \cdot \Delta T$$

$$P = \frac{Q}{s} = \frac{3 \cdot 1000 \cdot 4,2 \cdot 30}{60} = \frac{3 \cdot 30 \cdot 4,2 \cdot 1000}{3 \cdot 2 \cdot 10} = 6300 \text{ W}$$

$$P = 6300 \text{ W} \rightarrow P = V \cdot I \rightarrow I = \frac{P}{V} = \frac{6300}{220} \approx 28,6$$

$\rightarrow$ então disjuntor de 30 A $\rightarrow$ alternativa "b"